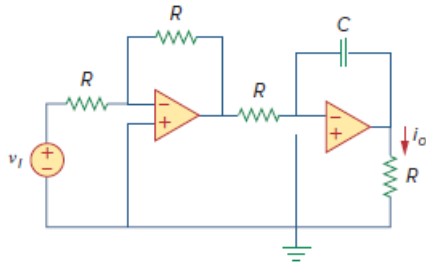


Problem

A gyrator is a device for simulating an inductor in a network. A basic gyrator circuit is shown in Fig. 16.113. By finding $V_i(s)/I_o(s)$, show that the inductance produced by the gyrator is $L = CR^2$.

Figure 16.113:



Step-by-step solution

Step 1 of 4

Consider a gyrator circuit:

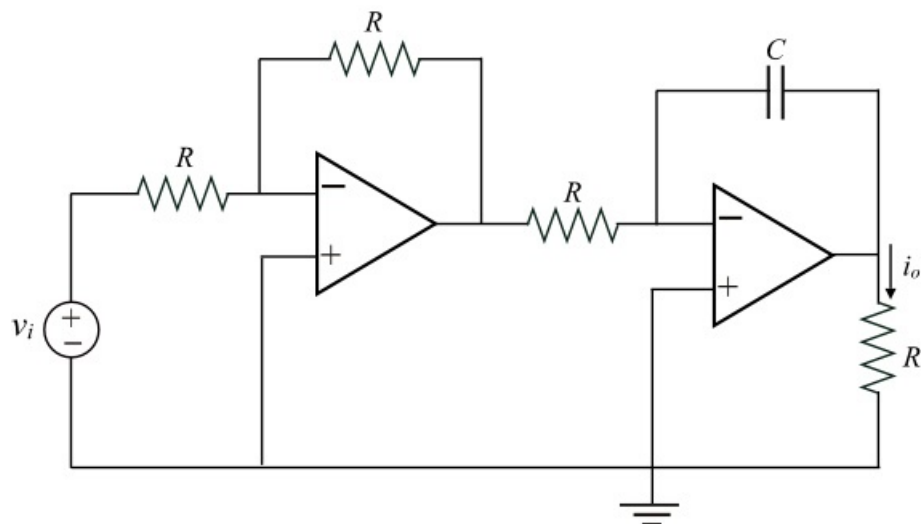


Figure 1

Step 2 of 4

Redraw the circuit in Figure 1 in s-domain.

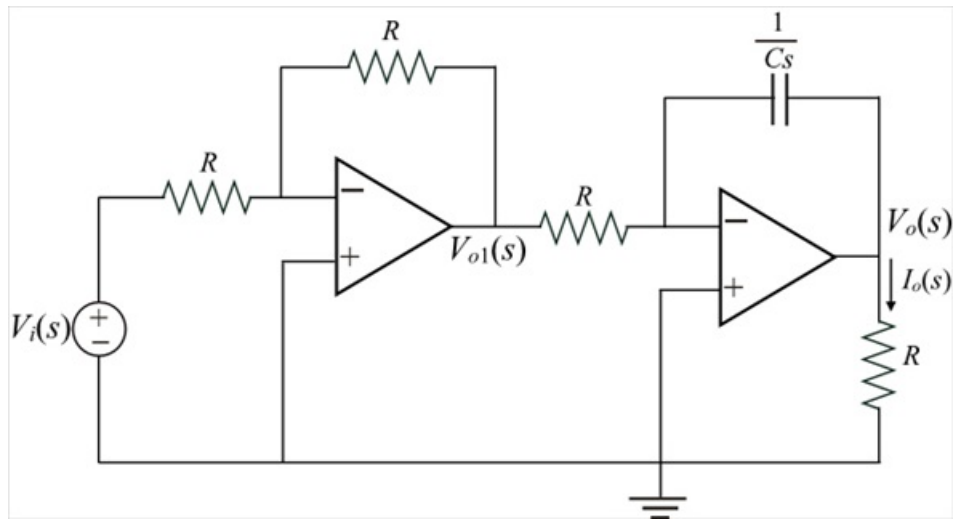


Figure 2

Step 3 of 4

Apply Kirchhoff's current law at the inverting terminal of the first op amp in s-domain.

$$\frac{V_i(s) - 0}{R} = \frac{0 - V_{o1}(s)}{R}$$

$$V_{o1}(s) = -V_i(s)$$

Apply Kirchhoff's current law at the inverting terminal of the second op amp in s-domain.

$$\frac{V_{o1}(s) - 0}{R} = \frac{0 - V_o(s)}{\left(\frac{1}{Cs}\right)}$$

$$V_o(s) = \frac{-V_{o1}(s)}{RCs}$$

Substitute $-V_i(s)$ for $V_{o1}(s)$.

$$V_o(s) = \frac{-(-V_i(s))}{RCs}$$

$$= \frac{V_i(s)}{RCs}$$

Step 4 of 4

Write the expression for $I_o(s)$ using Ohm's law.

$$\begin{aligned} I_o(s) &= \frac{V_o(s)}{R} \\ &= \frac{\left(\frac{V_i(s)}{RCs} \right)}{R} \\ &= \frac{V_i(s)}{R^2Cs} \end{aligned}$$

Rearrange the equation to find the expression for $\frac{V_i(s)}{I_o(s)}$.

$$\begin{aligned} \Rightarrow R^2Cs i_o(s) &= V_i(s) \\ \Rightarrow \frac{V_i(s)}{i_o(s)} &= R^2Cs \end{aligned}$$

Thus, the expression for $\frac{V_i(s)}{I_o(s)}$ is $\boxed{R^2Cs}$.

The ratio, $\frac{V_i(s)}{I_o(s)}$ is equal to inductive reactance of the network.

The inductive reactance in s-domain is sL .

Equate the value of the ratio, R^2Cs to sL .

$$R^2Cs = sL$$

$$L = R^2C$$

Hence, the inductance produced by the gyrator is $\boxed{R^2C}$.